

A02 SMOKE-FREE ENVIRONMENT | P

Intent : Deter smoking, minimize occupant exposure to secondhand smoke and reduce smoke pollution.

Summary : This WELL feature requires projects to ban indoor smoking and ban or restrict outdoor smoking within its boundaries.

Issue : Exposure to tobacco smoke has continued to detrimentally affect the health of both smokers and those exposed to secondhand smoke. Ingredients in cigarettes form over 7,000 compounds when burned, of which at least 69 are known to be carcinogenic.¹ As a result, the average life expectancy of a smoker is 10 years less than that of a nonsmoker.² Furthermore, secondhand smoke exposes nonsmokers to the same contaminants, increasing the number of people subject to health risks from smoking. The thirdhand smoke (residual chemicals left on indoor surfaces by tobacco smoke) clings to walls, furniture, clothing, bedding, carpets and other surfaces, long after smoking has occurred. Emerging evidence suggests that there are serious health consequences associated with exposure to thirdhand tobacco smoke,³ secondhand marijuana smoke and emissions from e-cigarettes.^{4,5} Health issues associated with tobacco smoke include asthma attacks, respiratory infections, coronary heart disease, stroke, lung cancer and sudden infant death syndrome.⁶ Despite these adverse impacts on health, 80% of the global population lives in countries not protected by 100% smoke-free regulations.⁷

Solutions : There is no safe and acceptable level of cigarette smoke exposure.⁸ Therefore, the only way to protect people from secondhand and thirdhand smoke is to implement a 100% smoke-free environment.⁸⁻¹⁰ In order to prevent intrusion of cigarette smoke from the outdoors, projects must also take steps to ensure that smoking is not allowed in the vicinity of building entrances, operable windows and building air intakes.¹¹

PART 1 PROHIBIT INDOOR SMOKING

For All Spaces:

The following requirement is met:

- a. Smoking and the use of e-cigarettes is prohibited in interior spaces within the project boundary.¹²

PART 2 PROHIBIT OUTDOOR SMOKING

For All Spaces:

Option 1: No smoking signage

The following requirements are met:

- a.
 - Clear and visible permanent signage prohibiting smoking and vaping is located within 10 ft of all functional building entrances, operable windows and building air intakes that open to any occupiable outdoor area.
 - Clear and visible permanent signage describing the hazards of smoking is located in all outdoor areas designated for smoking and vaping.

OR -----

Option 2: No applicable outdoor spaces

The following are not present within the project boundary:

- a. Any occupiable outdoor areas (e.g., decks, patios, balconies, rooftops, walkways)
- b. Functional building entrances.

REFERENCES

1. National Cancer Institute. Harms of Cigarette Smoking and Health Benefits of Quitting. <https://www.cancer.gov/about-cancer/causes-prevention/risk/tobacco/cessation-fact-sheet>.
2. Jha P, Ramasundarahettige C, Landsman V, et al. 21st-Century Hazards of Smoking and Benefits of Cessation in the United States. *N Engl J Med*. 2013;368(4):341-350. doi:10.1056/NEJMsa1211128
3. Matt GE, Quintana PJE, Destailats H, et al. Thirdhand tobacco smoke: Emerging evidence and arguments for a multidisciplinary research agenda. *Environ Health Perspect*. 2011;119(9):1218-1226. doi:10.1289/ehp.1103500
4. Wang X, Derakhshandeh R, Liu J, et al. One Minute of Marijuana Secondhand Smoke Exposure Substantially. *J Am Heart Assoc*. 2016;5(8):1-11. doi:10.1161/JAHA.116.003858
5. Mello S, Bigman CA, Sanders-Jackson A, Tan ASL. Perceived harm of secondhand electronic cigarette vapors and policy support to restrict public vaping: Results from a national survey of US adults. *Nicotine Tob Res*. 2016;18(5):686-693. doi:10.1093/ntr/ntv232
6. U.S. Department of Health and Human Services. The Health Consequences of Smoking: 50 Years of Progress: A Report of the Surgeon General. Rockville, MD; 2014. https://www.ncbi.nlm.nih.gov/books/NBK179276/pdf/Bookshelf_NBK179276.pdf.
7. World Health Organization. WHO Report on the Global Tobacco Epidemic, 2017 - Monitoring Tobacco Use and Prevention Policies. Geneva, Switzerland; 2017. <http://apps.who.int/iris/bitstream/handle/10665/255874/9789241512824-eng.pdf?sequence=1%0Ahttps://www.world-heart-federation.org/wp-content/uploads/2017/07/WHO-Report-on-the-global-tobacco-epidemic-2017-EMBARGOED.pdf%0Ahttp://apps.who.int/iris/bitstream/1>.
8. International Agency for Research on Cancer. Evaluating the Effectiveness of Smoke-Free Policies. Geneva, Switzerland; 2009. <https://www.iarc.fr/wp-content/uploads/2018/07/handbook13.pdf>.

9. Gan Q, Hammond SK, Jiang Y, Yang Y, Hu TW. Effectiveness of a smoke-free policy in lowering secondhand smoke concentrations in offices in China. *J Occup Environ Med*. 2008;50(5):570-575. doi:10.1097/JOM.0b013e3181638640
10. MacNaughton P, Adamkiewicz G, Arku RE, Vallarino J, Levy DE. The impact of a smoke-free policy on environmental tobacco smoke exposure in public housing developments. *Sci Total Environ*. 2016;557-558:676-680. doi:10.1016/j.scitotenv.2016.03.110
11. Kaufman P, Zhang B, Bondy SJ, Klepeis N, Ferrence R. Not just "a few wisps": Real-time measurement of tobacco smoke at entrances to office buildings. *Tob Control*. 2011;20(3):212-218. doi:10.1136/tc.2010.041277
12. New York State Department of Health. NYS Public Health Law, Article 13-E: Regulation of Smoking in Public and Work Places. New York, NY; 2014. <https://www.nysenate.gov/legislation/laws/PBH/A13-E>.
13. U.S. Green Building Council. LEED BD+C: New Construction | v4 - LEED v4. 2018. https://www.usgbc.org/sites/default/files/LEED_v4_BDC_07.2.18_current.pdf.
14. Hammond D, Fong GT, McNeill A, Borland R, Cummings KM. Effectiveness of cigarette warning labels in informing smokers about the risks of smoking: Findings from the International Tobacco Control (ITC) Four Country Survey. *Tob Control*. 2006;15(SUPPL. 3):iii19-25. doi:10.1136/tc.2005.012294